

A MORALE BOOSTER

The successful launch of GSLV-F04 proves that last year's failure was just an aberration and ISRO is back with a bang

■ By Raj Chengappa

This was the 25th Indian rocket launch that G. Madhavan Nair was involved with, but the Indian Space Research Organisation (ISRO) chairman looked far from relaxed. Four days before the scheduled lift-off of ISRO's most powerful space launcher, the Geosynchronous Satellite Launch Vehicle (GSLV), unseasonal squalls hampered work forcing the team to postpone the launch by a day.

Weighing on Nair's mind was last year's ignominious failure when a minute after the lift-off, GSLV-F02 had veered off course and broken up in midair. In just 62 seconds, Rs 300-crore worth of rocketry and satellites, that took two years to build, had turned into a massive fireball. Coming after an incredible run of 12 successful launches, the setback in space again put a big question mark on ISRO's capabilities. Especially because it was only the second operational launch of GSLV, a rocket ISRO hopes to market worldwide as a launcher for medium-sized communication satellites.

Typical of ISRO's culture, it admitted failure and vowed to set it right as quickly as possible. A failure analysis committee was constituted a day after the launch and its members spent over 100 hours poring over flight data and manufacturing documents apart from simulating tests. They found that the primary cause of failure was the sudden loss of thrust in one of the four strap-on motors that power the first stage of the rocket (*see chart*). That rapidly resulted in a significant reduction in control of the vehicle. As the errors built up after just 62 seconds of its flight, the aerodynamic pressure far exceeded the vehicle's design capability and it broke up. Having determined

SUCCESS: GSLV-F04 rocket lifts off from Sriharikota on September 2

